INCH-POUND

MIL-PRF-15733/39K
14 April 2009
SUPERSEDING
MIL-PRF-15733/39J
27 February 2003

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE,
HERMETICALLY SEALED, STYLE FL91

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PINs M15733/39-0007, -0008, -0011, -0013, are inactive for new design as of 19 August 1981. PINs M15733/39-0001, -0002, -0003, -0004, -0005, -0006, -0010, -0016, -0017, and -0018 are inactive for new design after 14 April 2009. For new design, see table II for substitution data.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification sheet and MIL-PRF-15733.

*

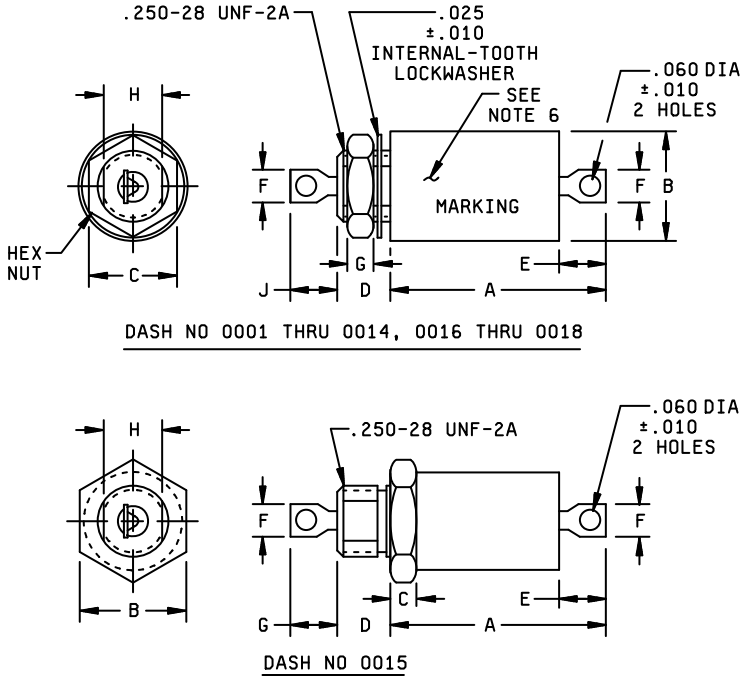


FIGURE 1. Case dimensions and circuit diagrams.

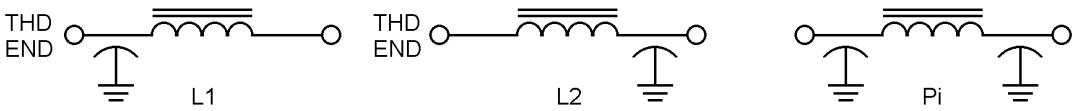
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Dash number	A Max	B Max	C Min	C Max	D Min	D Max	E Min	E Max	F Min	F Max	G Min	G Max	H Min	H Max	J Max	Maximum weight (grams)
0001 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0002 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0003 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0004 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0005 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0006 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0007 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0008 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0009	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0010 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0011 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0012	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.177 (4.50)	.197 (5.00)	.140 (3.56)	.190 (4.83)	.083 (2.11)	.130 (3.30)	.088 (2.24)	.098 (2.49)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0013 1/	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.302 (7.67)	.322 (8.18)	---	.190 (4.83)	---	.115 (2.92)	.091 (2.31)	.096 (2.44)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0014	.760 (19.30)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.302 (7.67)	.322 (8.18)	---	.190 (4.83)	---	.115 (2.92)	.091 (2.31)	.096 (2.44)	.190 (4.83)	.210 (5.33)	.150 (3.81)	28.4
0015	.509 (12.93)	.385 (9.78)	.120 (3.05)	.130 (3.30)	.302 (7.67)	.322 (8.18)	---	.190 (4.83)	---	.115 (2.92)	---	1.50 (3.81)	.190 (4.83)	.210 (5.33)	---	28.4
0016 1/	.80 (20.3)	.416 (10.57)	---	---	.180 (4.57)	.200 (5.08)	---	.357 (9.07)	.099 (2.51)	.119 (3.02)	---	---	.190 (4.83)	.210 (5.33)	---	10.0
0017 1/	.80 (20.3)	.416 (10.57)	---	---	.180 (4.57)	.200 (5.08)	---	.357 (9.07)	.099 (2.51)	.119 (3.02)	---	---	.190 (4.83)	.210 (5.33)	---	10.0
0018 1/	.80 (20.3)	.416 (10.57)	---	---	.180 (4.57)	.200 (5.08)	---	.357 (9.07)	.099 (2.51)	.119 (3.02)	---	---	.190 (4.83)	.210 (5.33)	---	10.0

1/ Inactive for new design.

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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CIRCUIT DIAGRAM

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.
4. Circuit diagram is for information only.
5. Filters shall be supplied with mounting hardware.
6. Terminal identification (nonsymmetrical filters): The case shall be marked at the threaded end of the filter, with the symbol “C” or the symbol “L”, as follows:

Circuit	Symbol
L_1	C
L_2	L

7. Recommended mounting torque: 48 in-oz.

FIGURE 1. Case dimensions and circuit diagrams – Continued.

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TABLE I. Electrical characteristics.

Rated voltage (volts dc)	Circuit diagram	Max current (amps)	Max voltage drop (volts)	Capacitance µF GMV	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/																Dash number
					AT +25°C								AT -55°C and +125°C								
					30 kHz	100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	30 kHz	100 kHz	150 kHz	300 kHz	1 MHz	10 MHz	100 MHz	1 GHz	
100	L_1	.5	1.0	.66	6	---	28	40	60	60	60	70	4	---	26	38	58	58	60	70	0001 2/
100	L_2	.5	1.0	.66	6	---	28	40	60	60	60	70	4	---	26	38	58	58	60	70	0002 2/
100	L_1	1.0	.25	.45	6	---	20	30	50	60	60	70	4	---	18	28	48	58	60	70	0003 2/
100	L_2	1.0	.25	.45	6	---	20	30	50	60	60	70	4	---	18	28	48	58	60	70	0004 2/
100	L_1	2.0	.14	.45	6	---	20	26	40	60	60	70	4	---	18	24	38	58	60	70	0005 2/
100	L_2	2.0	.14	.45	6	---	20	26	40	60	60	70	4	---	18	24	38	58	60	70	0006 2/
100	L_1	5.0	.075	.45	6	---	20	26	35	60	60	70	4	---	18	24	33	58	60	70	0007 2/
100	L_2	5.0	.075	.45	6	---	20	26	35	60	60	70	4	---	18	24	33	58	60	70	0008 2/
100	L_1	10.0	.03	.45	6	---	20	26	35	56	60	70	4	---	18	24	33	54	60	70	0009
100	L_2	10.0	.03	.45	6	---	20	26	35	56	60	70	4	---	18	24	33	54	60	70	0010 2/
100	P_i	10.0	.03	.99	9	---	24	29	40	70	70	70	7	---	22	27	38	68	70	70	0011 2/
100	P_i	10.0	.03	.99	9	---	24	29	40	70	70	70	7	---	22	27	38	68	70	70	0012
100	P_i	10.0	.03	1.5	9	---	24	29	40	70	70	70	7	---	22	27	38	68	70	70	0015
100	P_i	.25	.38	1.5	---	58	---	76	80	80	80	80	---	56	---	74	78	78	80	80	0016 2/
100	P_i	3.0	.18	1.5	---	14	---	41	76	80	80	80	---	12	---	39	74	78	80	80	0017 2/
100	P_i	5.0	.20	1.5	---	---	---	28	68	80	80	80	---	---	---	26	66	78	80	80	0018 2/
50	L_2	10.0	.03	1.4	14	---	28	34	44	52	70	70	12	---	26	32	42	50	70	70	0013 2/
50	L_1	10.0	.03	1.4	14	---	28	34	44	52	70	70	12	---	26	32	42	50	70	70	0014

1/ Insertion loss measurements shall be made under full load over the frequency range of 100 kHz to 10 MHz. Insertion loss measurements above or below this frequency range shall be made under no load.

2/ Inactive for new design.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: See figure 1.

Case: Metal.

* Case and mounting hardware finish: In accordance with MIL-PRF-15733. Pure tin finish is prohibited as specified in MIL-PRF-15733.

Seal: Metal-to-glass.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: See table I.

Insertion loss: In accordance with MIL-PRF-15733 and table I.

Seal: In accordance with MIL-PRF-15733.

Capacitance to ground: In accordance with MIL-PRF-15733 and table I.

Dissipation factor: 3.0 percent maximum.

Temperature rise: 25°C, maximum.

Barometric pressure (reduced): In accordance with MIL-PRF-15733 and method 105, MIL-STD-202, test condition D (100,000 ft) for -0001 through -0010; Test condition C (70,000 ft) for -0011 through -0015; and test condition B (50,000 ft) for -0016 through -0018.

Insulation resistance: In accordance with MIL-PRF-15733 and the following:

At 25°C the insulation resistance shall not be less than 1000 megohms.

At +125°C the insulation resistance shall not be less than 100 megohms.

Voltage drop: See table I.

Terminal strength: In accordance with MIL-PRF15733 and method 211, MIL-STD-202; test condition A (pull).

Force: 5 pounds.

Salt atmosphere (corrosion): In accordance with MIL-PRF-15733 and method 101, MIL-STD-202; test condition A.

Shock (specified pulse): In accordance with MIL-PRF-15733 and method 213, MIL-STD-202. The following details apply:

Test condition I: (-0001 through -0014 and -0016 through -0018).

Test condition A: (-0015).

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Vibration, high frequency: In accordance with MIL-PRF-15733 and method 204, MIL-STD-202; test condition D (20G). The following detail shall apply:

Temperature conditioning: Prior to vibration, filters shall be heated in an oven to +125°C plus one-half the temperature rise. Vibration in each plane shall begin within 5 minutes after removal of filters from the oven.

Life: In accordance with MIL-PRF-15733 and method 108, MIL-STD-202; test condition D (1,000 hrs) for qualification; test condition B (250 hrs) for group C inspection.

Part or Identifying Number (PIN): M15733/39-(dash number from table I).

Extension of qualification:

Qualification testing and approval to M15733/39-0014 shall be sufficient to grant qualification approval to M15733/39-0013 and M15733/39-0014.

Qualification testing and approval to M15733/39-0012 and M15733/39-0016 shall be sufficient to grant qualification approval to M15733/39-0001 through -0012 and M15733/39-0015 through -0018.

* TABLE II. Substitution data.

Inactive PIN	For new design use	
	Superseding specification	Superseding PIN
M15733/39-0001	MIL-PRF-28861/3	M28861/03-007TB
M15733/39-0002	MIL-PRF-28861/3	M28861/03-008TB
M15733/39-0003	MIL-PRF-28861/2	M28861/02-004TB
M15733/39-0004	MIL-PRF-28861/2	M28861/02-005TB
M15733/39-0005	MIL-PRF-28861/2	M28861/02-007TB
M15733/39-0006	MIL-PRF-28861/2	M28861/02-008TB
M15733/39-0007	MIL-PRF-28861/2	M28861/02-010TB
M15733/39-0008	MIL-PRF-28861/2	M28861/02-011TB
M15733/39-0010	MIL-PRF-28861/1	M28861/01-005TB
M15733/39-0011	MIL-PRF-15733/39	M15733/39-0012
M15733/39-0013	MIL-PRF-28861/1	M28861/01-001TB
M15733/39-0016	MIL-PRF-28861/2	M28861/02-003TB
M15733/39-0017	MIL-PRF-28861/2	M28861/02-009TB
M15733/39-0018	MIL-PRF-28861/2	M28861/02-012TB

NOTES:

* Referenced documents. In addition to MIL-PRF-15733, this specification sheet references the following documents:

MIL-PRF-28861/1
MIL-PRF-28861/2
MIL-PRF-28861/3
MIL-STD-202
MIL-STD-220

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5915-2009-003)

Review activities:
Army - AT, AV
Navy - AS, CG, MC, OS, SH
Air Force – 99

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